

Transdisciplinary Platform Assets – A Boundary Object Perspective

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Abstract. In product platforms, reusable assets play an important role in supporting the realization of new products and systems. Asset governance is key to ensure their relevance and useability. Assets can have different aims for different stakeholders, where design engineers might use design guidelines as platform assets within their organizational function, while other assets aim to support transdisciplinary collaboration, such as the production preparation process. Although assets used within functions have received previous attention, the literature has not yet presented a clear path forward for how the transdisciplinary perspective on platform asset governance can be supported. To shed light on this topic, we therefore introduce the concept of Transdisciplinary Platform Assets. We do this by conceptualizing these Transdisciplinary Platform Assets from a boundary object perspective. Boundary objects can enable transdisciplinary collaboration and integration of knowledge across diverse communities. By viewing assets from a boundary object perspective, we can illuminate how assets can be tailored for transdisciplinary usage; subsequently becoming more effective. In this paper we put forward a way to classify assets based on the width of the transdisciplinary gap between the users. The concept is further illuminated in its implications through an illustrative discussion on A3 templates. We conclude by highlighting potential paths for future research and potential industrial use cases.

Keywords. Product platform, boundary objects, platform assets, product realization, transdisciplinary engineering

Introduction

The realization of new products is an imperative task for manufacturing companies, where the products need to be developed in a timely manner and contain the desired features at the right price. A subsequent effect of the multitude of aspect affecting the success of the product is the need for transdisciplinary interaction and coordination across functions such as research and development (R&D), production, and marketing [1]. The advent of the product platform concept can be seen as a potential solution to some of the challenges at hand, where modularization of the product, as well as management of other platform assets can reduce the development time through reuse. Through the past research on platforms, several different conceptualizations has been presented [2]. In this research, we adhere to the view that platforms consist of reusable assets required to realize a company's offering to the market [3]. Here, the assets are a collection of the artefacts, knowledge and activities used in a company [4].

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Although product platform approaches have been seen to support manufacturing companies, there are still challenges pertaining to the transdisciplinary nature of product realization. For example, when the different organizational functions work together and solve problems, they start to form communities of practice which deepen the knowledge within their domain [5]. However, an effect of this is that the very specialization that can be seen as a necessity for the development of innovative new products subsequently may cause challenges in the creation of common understanding across functions [6]. Thus, a product platform and its assets should not only accommodate for the work *within*, but also *across* functions.

For a product platform to enable transdisciplinary interactions, one way can be to look at its constituent components that are utilized during product realization; the assets. However, a limitation in the literature as of yet is that there is no clear distinction between the diverging purposes that the assets may have. Whereas some may serve the purpose of enabling engineers to design products from coherent design principles and thus support intradisciplinary work, some should serve to facilitate transdisciplinary interactions. These assets subsequently have different demands on them in terms of their usage and further governance. To support and shed lighter on the matter, we therefore propose the concept of Transdisciplinary Platform Assets (TPA), which may help us to gain deeper insights into how the different assets can be governed for successful product realization.

In order to introduce and discern what TPA are, a number of questions can be posed. For instance, the distinction between TPA and intradisciplinary platform assets is reliant on clearly defining what they are and how they differ. Based around the notion that these assets differ; a further area of interest becomes that of platform governance and how the assets should be managed over time.

As TPA aims to facilitate transdisciplinary interactions, a meaningful lens to apply is that of boundary objects. Boundary objects facilitate the creation of transdisciplinary understanding when they possess desired properties and are used in a suitable situation [7]. In this paper we therefore leverage and synthesize the literature on platform assets and boundary objects to shed light on the aforementioned questions.

1. Product platforms, assets, and boundary objects

Scientific literature often present how product platforms can contribute to cost savings, shorter time-to-market, and improved product quality. Moreover, different definitions, approaches, and methods for the creation of product platforms are well explored in the scientific literature. Rather than describing the reported merits of platforms, this section is focused on the elements forming a platform, and how these may be used in different situations.

1.1. Product Platform Assets

A product platform can be defined as a collection of four categories of assets: *Components, Processes, Knowledge, People and relationships* [4]. This generic definition has been further elaborated into the *Design Platform*, through studies of the practices of different industries [3]. Here, eight classes of objects have been identified that can be considered a “toolbox” for the development teams, forming the backbone of

a design platform. In an industrial case study [8], the authors explicate the role of the design platform as an explicit, shared, and clearly defined overarching model for organizing design assets, which can be continually augmented to support a dynamic, evolving platform. Typical assets utilized in practical applications within a mechanical design department are outlined in [8]. This work also highlights that crucial tools and practices often lack documentation or formalization, rendering them outside the realm of the design platform and inaccessible to the organization. Specifically, knowledge and learning are identified as essential prerequisites for synthesizing new products and should therefore be regarded as significant assets. To optimize their effectiveness, they must be tailored to suit the intended user [9].

Platforms undergo evolutionary processes over time, delineated into three fundamental life cycle phases: preparation, execution, and maintenance, as defined in [10]. These life cycle phases significantly impact the requirements placed on knowledge assets, as the degree of novelty influences their applicability and internalization by users, contingent upon the stage of novelty within the development cycle [11].

1.2. Boundary objects

In knowledge intensive product realization, there is a need for tools promoting cross-disciplinary and cross-cultural integration. Boundary objects are concepts, artifacts, processes, and other entities that serve as tools to foster coherence in understanding, bridging intellectual boundaries among individuals from diverse backgrounds. Since the concept of boundary objects was first incepted in 1989 [12], it has received ample attention in multiple disciplines and areas of application.

Something that has challenged researchers is what features that makes something a boundary object or not, as it is not certain that an object that once functioned as a boundary object will forever do so [6]. This can be linked to the situational nature of boundary objects, which implies that the usage of a specific object and its role is closely linked to the situation in which it is used. To shed light on the underlying mechanics of boundary objects, recent literature has distinguished different aspects that affects how an object function as a boundary object [7]. Notably, two categories of aspects have been identified, where the first relates to the object itself. These object specific aspects are referred to as properties [13], and can for instance relate to the object's visualization [14]. To account for the situation in which the object is used, the distinction of situational aspects has been proposed [7], which can for instance be linked to the space in which the object is used [15]. The interplay between the situational aspects and properties has been seen to play a role, where different aspects may affect one another and subsequently either support or hinder the boundary object [7].

2. Outlining Transdisciplinary Platform Assets (TPA)

In order to define what TPA entails, we must first define what they aim to support. As previously argued in this paper, a key objective for TPA is to support transdisciplinary collaboration. Coming from this, two different parts may be discerned, where one significant part of this is that of transdisciplinarity, and the second part related to collaboration. In relation to the first aspect of transdisciplinarity, we argue that disciplines can exist on different levels of abstraction, which subsequently brings with

them different transdisciplinary distances. Figure 1 portrays an example of this discussion, where we exemplify how a manufacturing company organization may look. To exemplify in relation to product realization, Ulrich, et al. [1] defines that product realization typically involves the functions of product design, manufacturing, and marketing. These can in themselves be seen as different disciplines. However, within these disciplines, sub-disciplines may also exist. Product design can for instance also include engineers who work with mechanics, software, or electronics. On an even lower level of abstraction, a subset of mechanical engineering can be those individuals having their competence mainly within geometrical modelling in computer aided design (CAD) or solid mechanics using finite element analysis (FEM). These subsets can be argued to in themselves constitute different disciplines, and as such, demonstrate different levels of transdisciplinary distance between them. This is exemplified in figure 1, where we can distinguish three different levels of transdisciplinary distance. For instance, the distance between CAD and FEM engineers is lower than that of mechanical and electronical engineering since CAD and FEM belong to a lower-level disciplinary denominator. In the figure we exemplify three levels, although there can exist an even higher number of levels, who are not directly involved with the product realization.

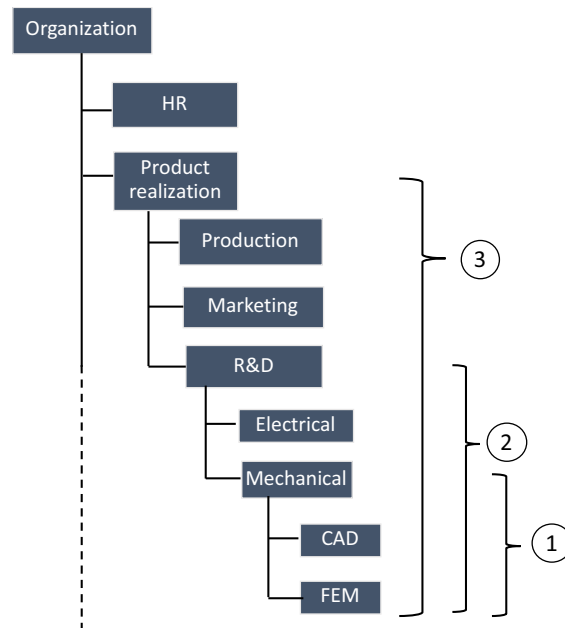


Figure 1. Transdisciplinary distances in an organization. Note that a difference in interdisciplinary knowledge domains can be argued to in themselves constitute different disciplines and thus create different levels of transdisciplinary distance.

Another implication of transdisciplinary distance is that of the hierarchical distance, where managers on a higher level in the company can be further away disciplinary, as compared to engineers working together in a lower-level disciplinary group. This can be further connected to the experience of individuals, where an experienced individual has different needs from an asset as compared to a less experienced one [16]. For instance,

senior engineers may be less dependent on assets such as design guidelines since they generally possess a larger transdisciplinary network and a higher degree of tacit knowledge based on their previous experiences [16]. Familiarity and experience can thus further be seen as playing a role in the transdisciplinary complexity which the TPA should aim to support. As seen in the literature, novelty plays a major role in transdisciplinary interactions, where the amount of novelty drives complexity in the boundaries needed to be crossed [6]. However, as we argue, transdisciplinary complexity does not only stem from novelty, but also from the transdisciplinary distance between the disciplines. As such, we present an elaboration on Carlile's previous work by adding the dimension of transdisciplinary distance as discussed earlier. A synthesis of the transdisciplinary distance and the novelty can be seen in figure 2, where the drivers for transdisciplinary complexity stem from the aforementioned components.

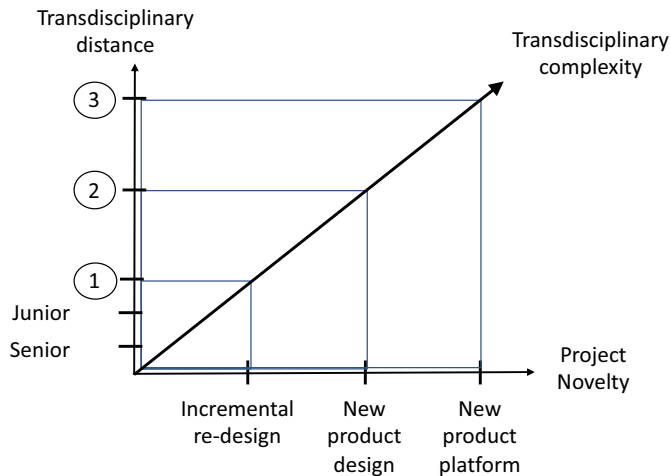


Figure 2. The transdisciplinary complexity can be illustrated through the two drivers' transdisciplinary distance and novelty.

In Figure 2, we have outlined the distance between junior and senior individuals within a discipline as a factor to be considered when discussing transdisciplinary complexity. Based on the distinction of the transdisciplinary complexity at hand, different types of assets may serve better as TPA, which leads us back to the topic of collaboration. As product realization is a knowledge intensive endeavor, the purposeful integration of specialized and complementary knowledge becomes imperative [17]. Here we can leverage boundary object theory as a way to gain further insights into what types of assets may serve well as TPA. As noted by Carlile [6], the higher the amount of novelty at hand, different types of objects may work differently well to facilitate the integration of knowledge and creation of common understanding across disciplines. Thus, a platform should be able to coordinate different types of TPA which can function as boundary objects. As such, including TPA in a platform approach serves as a way to

coordinate the TPA and further facilitates transdisciplinary interactions. Another benefit of TPA is further through their reuse. As Carlile (2002) notes that novelty is a driver for complexity, we must also note that novelty may come not only from the innovation scope, but also from the use of the assets. As assets are reused, the novelty from the users goes down; they gain familiarity with said asset and its purpose. The concept of TPA thus further supports transdisciplinary endeavors since it paves the way for decreased novelty in asset use.

3. Governance of Transdisciplinary Platform Assets

By leveraging boundary object theory, we can shed further light on the demands put on TPA governance. For instance, the asset itself plays a major role in whether it facilitates transdisciplinary interaction or not. This can be linked to the different types of properties [13], that an asset may possess. As noted, the higher levels of complexity, the higher the demands are on the asset, where different properties become of high essence if it is to function as a boundary object [7]. However, higher transdisciplinary complexity does not only require different types of assets, but also puts higher demands on the situation in which they are used, where different situational aspects play a role [7].

Connecting the transdisciplinary complexity to the properties and situational aspects, some guidance related to TPA governance can be discerned. Looking at boundary object theory, low amounts of transdisciplinary complexity advocates assets that serve as repositories of information, from which individuals can access codified knowledge. On the other hand, higher complexity requires an object which is more flexible and open for higher amounts of interpretative flexibility [7]. Higher complexity further puts higher requirements on the situation in which the TPA is used. For example, a broker can support the TPA's function as a boundary object since the broker is able to fill in contextual gaps that the TPA in itself might have [18]. As such, there is a complementing role between the situation in which the TPA is used and whether it functions as a boundary object or not. Thus, a further distinction between TPA and intra-disciplinary assets is that TPA not only puts requirements on the asset itself, but also on the situation in which it is used. This in return becomes an important part related to the governance of the asset, where TPA used for higher transdisciplinary complexity also needs to have additional information regarding its usage prescribed.

By combining the earlier discussion on transdisciplinary complexity laid out in figure 1, companies can position their assets and classify them based on the transdisciplinary complexity they are meant to cater to, thereby making the reuse of assets more efficient. The depiction of transdisciplinary complexity can thus support the governance of TPA by discerning the assets currently at hand, but also the potential gaps that may exist in terms of TPA.

4. Illustrative case

To illustrate the implications of TPA, we here present an illustrative case on A3 templates as TPA. As noted earlier, higher degrees of novelty and complexity require an object which is more flexible [7, 19]. An implication of this on A3 templates can thus be linked to the way the A3 template is designed and structured. For example, an A3 template filled with lessons learned has less flexibility since it is concrete and rigid [7], but on the other

hand works well as a repository for knowledge [6]. In situations with lower levels of complexity, the A3 templates with lessons learned can as such be a relevant TPA, whereas in situations with higher complexity it might not. Thus, to cater to situations with higher amount of complexity, an A3 template which is more open ended, potentially containing questions for discussion, leaves more room for interpretative flexibility [12, 19], and can thus be a suitable TPA for situations with higher degrees of transdisciplinary complexity. In situations with even higher amounts of transdisciplinary complexity, it might be required to have a broker present who can facilitate the transdisciplinary interaction. A visual depiction of this is presented in Figure 3 below.

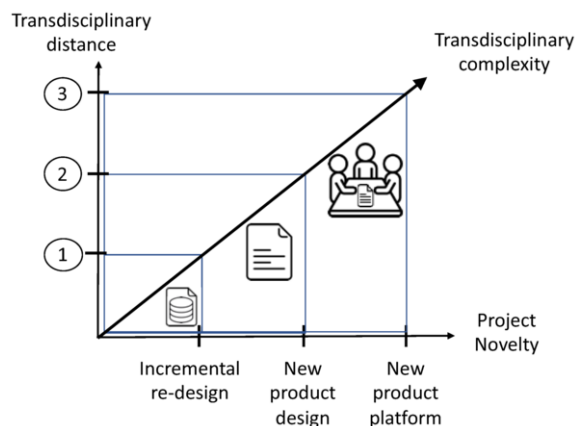


Figure 3. Examples of platform assets for different transdisciplinary complexity. For low levels, codified information in repositories is sufficient, for medium levels, higher flexibility is needed and for high levels of complexity, co-location and support from a broker is advised.

5. Discussion and conclusion

In this paper we have outlined the concept of TPA and discussed its implications in relation to both transdisciplinary interactions, as well as an integral part of platform governance. By distinguishing assets which serve as TPA we can provide more insights into product platforms and their assets, which further provides guidance into the governance of assets. For instance, due to their disparate backgrounds and experiences, junior and senior engineers within a discipline will exhibit an intra-disciplinary distance and may therefore require more flexible assets for their collaboration.

Our contributions to literature can be seen in multiple ways. First, we have defined and elaborated on the constituent components of transdisciplinary complexity, where we put forward a way to distinguish different levels of transdisciplinary distances and how they subsequently result in different amounts of transdisciplinary complexity. The distinction and definition of transdisciplinary complexity thus adds to the field of transdisciplinary engineering [20] by elaborating on the implication of transdisciplinary distances coupled with novelty. Since we posit that TPA governance differs from governance of intradisciplinary assets, we further add to the literature on product platforms, where finer granularity and insights can be gained from out conceptualization of TPA. Since the paper has leveraged boundary object theory, we have further reduced the gap between boundary object and platform theory by synthesizing.

To support the further development of the findings of this paper, we can define a set of auspicious paths for future research. For the concept of transdisciplinary complexity, a future avenue could be to further elaborate on the constituent components that adds to transdisciplinary complexity. Furthermore, as TPA induces new requirements on governance, further exploration into the integration of platform governance is needed, both from a theoretical as well as a practical standpoint. Finally, we propose empirical research work that apply our concepts to assess their practical contributions.

Acknowledgement

We would like to thank the participating company, the respondents, the IDEAL project group, the Swedish Knowledge Foundation, and the SPARK environment at Jönköping university for their support.

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